

UTSJOKI KEVO, Finland

WMO: 028050

Lat: 69.756N

Lon: 27.007E

Elev: 106

StdP: 100.05

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.6	-31.6	-37.8	0.1	-34.4	-34.5	0.1	-31.5	13.6	-3.8	10.8	-4.1	1.4	160	0.760

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.0	24.3	16.2	21.9	15.2	19.4	14.0	17.3	22.5	15.9	20.6	14.7	18.7	3.4	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.2	10.9	19.3	13.8	10.0	17.5	12.6	9.2	16.3	49.1	22.4	44.8	20.7	41.4	18.9	22.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.3	7.0	-37.2	28.0	3.3	2.3	-39.6	29.7	-41.5	31.0	-43.4	32.4	-45.8	34.1		
(5)			-37.2	19.0	3.3	1.5	-39.6	20.1	-41.5	21.0	-43.3	21.9	-45.7	23.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.9	-13.8	-13.3	-8.3	-2.2	3.9	9.6	13.5	11.2	6.6	-0.2	-7.8	-10.9	(6)
(7)		DBStd	11.15	8.97	8.63	6.66	5.06	3.81	3.75	3.63	3.39	3.42	4.91	7.02	8.17	(7)
(8)		HDD10.0	4216	738	653	567	367	193	52	8	26	112	318	533	649	(8)
(9)		HDD18.3	7034	997	886	825	617	446	263	157	223	353	576	783	907	(9)
(10)		CDD10.0	231	0	0	0	0	5	39	116	62	9	0	0	0	(10)
(11)		CDD18.3	8	0	0	0	0	0	1	6	2	0	0	0	0	(11)
(12)		CDH23.3	119	0	0	0	0	3	11	83	22	0	0	0	0	(12)
(13)		CDH26.7	23	0	0	0	0	1	1	18	3	0	0	0	0	(13)
(14)	Wind	WSAvg	2.9	2.7	2.6	3.0	2.9	3.3	3.7	3.2	2.7	2.8	2.7	2.5	2.7	(14)
(15)	Precipitation	PrecAvg	416	26	21	20	21	23	45	69	58	41	38	29	26	(15)
(16)		PrecMax	597	67	66	40	55	81	115	157	161	82	97	61	53	(16)
(17)		PrecMin	263	6	3	5	6	1	3	13	11	5	8	7	7	(17)
(18)		PrecStd	85	12	12	10	11	16	25	36	28	19	20	13	11	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.0	3.6	5.0	11.0	21.6	24.8	28.9	26.2	18.1	10.9	4.5	4.5	(19)
(20)			MCWB	1.6	2.0	2.6	6.2	12.1	15.4	17.8	17.6	13.1	8.9	3.6	2.6	(20)
(21)		2%	DB	1.1	1.8	3.1	8.1	16.6	21.8	25.6	22.5	15.7	9.3	2.7	2.0	(21)
(22)			MCWB	0.1	0.5	1.3	4.4	9.7	13.7	17.0	16.1	12.2	7.7	1.7	0.9	(22)
(23)		5%	DB	-0.8	0.1	1.5	5.9	13.2	18.9	22.9	19.7	13.7	7.5	1.7	0.7	(23)
(24)			MCWB	-1.6	-1.0	-0.1	3.0	8.1	12.5	15.8	14.7	11.2	6.2	1.0	-0.2	(24)
(25)		10%	DB	-2.4	-1.7	0.2	4.3	10.3	16.4	20.6	17.3	12.1	5.8	0.7	-0.5	(25)
(26)			MCWB	-3.1	-2.6	-1.2	2.0	6.5	11.3	15.1	13.5	10.1	4.7	0.1	-1.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.8	2.3	2.8	6.5	13.3	16.5	19.4	18.6	14.3	9.3	3.7	2.7	(27)
(28)			MCDB	2.9	3.5	4.7	10.4	20.7	21.8	25.4	23.8	17.2	10.6	4.4	4.3	(28)
(29)		2%	WB	0.1	0.6	1.3	4.6	10.3	14.8	18.0	16.7	12.7	7.8	2.0	1.1	(29)
(30)			MCDB	1.0	1.8	2.8	7.8	16.0	19.7	23.4	21.5	14.8	9.1	2.5	1.9	(30)
(31)		5%	WB	-1.5	-0.9	0.0	3.2	8.4	13.2	16.7	15.1	11.5	6.3	1.1	-0.1	(31)
(32)			MCDB	-0.8	0.1	1.3	5.5	12.6	17.8	21.8	18.9	13.4	7.4	1.6	0.7	(32)
(33)		10%	WB	-3.1	-2.5	-1.1	2.2	6.8	11.7	15.3	14.0	10.3	4.9	0.2	-1.3	(33)
(34)			MCDB	-2.4	-1.7	0.1	4.0	10.1	16.0	20.0	17.0	11.9	5.8	0.6	-0.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.8	8.9	10.2	9.5	8.0	8.3	9.0	8.7	7.3	5.2	7.0	8.4	(35)
(36)			MCDBR	7.5	6.8	7.2	9.5	14.1	13.1	14.4	13.6	10.2	5.9	5.2	6.6	(36)
(37)			MCWBR	7.1	6.3	6.0	6.5	8.1	6.5	6.8	7.3	6.8	4.7	4.8	6.2	(37)
(38)		5% WB	MCDBR	7.5	6.5	7.2	8.8	12.9	11.6	12.7	11.8	9.1	5.6	5.0	6.5	(38)
(39)			MCWBR	7.1	6.1	6.2	6.1	7.8	6.2	6.6	6.7	6.4	4.7	4.7	6.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.202	0.247	0.295	0.324	0.346	0.344	0.356	0.338	0.306	0.260	N/A	N/A	(40)	
(41)		taud	1.903	2.027	2.104	2.156	2.314	2.452	2.444	2.475	2.525	2.206	N/A	N/A	(41)	
(42)		Ebn at Noon	14	466	695	800	833	851	824	797	728	461	0	0	(42)	
(43)		Edh at Noon	3	42	80	105	102	93	90	78	56	34	0	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.01	0.35	1.71	3.57	4.68	4.74	4.37	3.02	1.57	0.48	0.00	0.00	(44)	
(45)		RadStd	0.00	0.03	0.11	0.21	0.49	0.43	0.48	0.29	0.14	0.06	0.00	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	+0.45	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page